

TAMBOV, Russian Federation

WMO: 279470

Lat: 52.800N

Lon: 41.333E

Elev: 421

StdP: 14.47

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.2	-9.4	-19.6	1.9	-14.6	-14.3	2.6	-8.9	23.0	27.4	20.7	24.9	3.6	350	0.578

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	89.9	68.0	86.1	67.1	82.8	65.8	70.9	84.8	69.3	82.0	67.7	79.4	8.2	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.4	98.6	75.8	64.8	93.2	74.5	63.3	88.5	73.0	34.9	85.1	33.6	82.0	32.3	79.6	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.7	17.4	15.6	-20.9	93.9	6.6	4.0	-25.7	96.7	-29.5	99.0	-33.2	101.2	-37.9	104.1		
(5)			-21.2	73.4	6.5	1.7	-25.9	74.6	-29.7	75.6	-33.3	76.6	-38.0	77.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.5	17.3	18.2	28.0	45.8	58.7	64.6	69.1	66.2	55.9	43.5	31.4	22.2		
	DBStd	20.89	12.99	12.82	9.16	8.95	8.72	7.21	6.33	7.23	7.75	8.25	9.66	11.84		
	HDD50	4483	1013	890	683	186	22	2	0	0	30	235	559	863		
	HDD65	8234	1478	1310	1147	577	232	96	29	73	287	668	1009	1328		
	CDD50	2126	0	0	0	59	293	440	592	502	206	33	1	0		
	CDD65	400	0	0	0	1	38	83	156	109	13	0	0	0		
	CDH74	4227	0	0	0	17	426	861	1561	1203	159	0	0	0		
	CDH80	1428	0	0	0	1	102	268	569	460	28	0	0	0		
	WSAvg	7.7	8.4	9.0	8.7	8.0	7.4	6.8	6.0	6.2	6.6	7.9	8.6	8.8		
	PrecAvg	20.8	1.5	1.3	1.1	1.1	1.8	2.8	2.2	1.9	2.0	1.9	1.7	1.6		
(15) Precipitation	PrecMax	29.8	2.7	2.8	2.4	3.9	5.7	6.1	5.2	4.6	5.1	4.1	4.3	3.3		
	PrecMin	13.4	0.7	0.2	0.1	0.2	0.5	1.1	0.4	0.3	0.1	0.7	0.1	0.4		
	PrecStd	4.3	0.5	0.7	0.6	0.8	1.3	1.2	1.2	1.1	1.3	0.9	1.1	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.5	38.3	55.4	76.5	87.3	91.0	96.2	97.6	83.7	69.3	54.2	45.3		
		MCWB	36.8	36.3	46.4	55.7	65.0	66.4	70.7	70.5	62.3	56.9	50.9	44.0		
	2%	DB	35.7	36.5	47.6	70.9	82.9	86.4	89.9	89.9	78.6	64.1	50.0	40.6		
		MCWB	34.1	34.8	41.4	54.1	63.0	67.1	69.0	67.9	61.0	54.9	47.9	38.9		
	5%	DB	34.3	35.1	43.0	66.3	79.0	83.0	86.1	84.7	74.0	59.3	46.7	37.2		
		MCWB	33.1	33.6	38.3	52.4	61.6	66.4	68.2	66.5	60.1	52.8	44.7	35.8		
	10%	DB	33.1	33.8	39.1	61.9	74.9	79.1	82.3	80.5	69.4	55.7	43.3	34.9		
		MCWB	32.0	32.5	36.0	50.2	59.7	64.6	66.8	64.9	58.1	50.9	41.6	33.7		
	0.4%	WB	37.3	37.1	46.8	58.1	67.7	71.7	73.0	72.6	65.9	60.7	51.5	43.8		
		MCDB	38.2	37.8	53.8	69.4	82.8	83.2	88.1	91.4	76.9	66.6	53.7	44.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	34.5	35.2	42.2	55.5	64.9	69.7	71.2	70.0	63.3	56.5	48.0	39.3		
		MCDB	35.2	36.0	47.0	68.0	78.7	81.7	85.4	84.0	73.1	61.1	49.8	40.3		
	5%	WB	33.3	33.9	38.5	53.3	62.8	67.9	69.7	68.2	61.3	53.7	44.9	36.0		
		MCDB	34.2	34.8	42.0	64.3	76.2	79.2	82.9	81.3	71.2	58.5	46.5	36.9		
	10%	WB	32.2	32.6	36.2	51.2	60.8	66.1	68.1	66.3	59.4	51.1	41.8	33.9		
		MCDB	33.3	33.8	38.8	60.7	72.4	76.4	79.7	77.5	68.1	55.2	43.1	34.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.4	12.0	13.1	18.6	20.7	19.1	20.0	21.1	18.6	13.1	8.9	9.2		
		MCDBR	6.6	7.9	17.3	25.6	25.7	25.0	25.2	27.7	25.9	19.7	11.8	6.9		
		MCWBR	6.7	7.4	11.5	12.4	10.8	10.0	9.3	10.4	11.7	12.2	9.9	6.4		
		MCDBR	6.6	7.7	14.9	22.7	23.4	21.7	22.3	23.4	21.7	16.9	10.4	7.1		
	5% WB	MCWBR	6.7	7.7	10.6	11.9	11.0	10.1	9.4	10.1	11.2	12.0	9.3	7.0		
(40) Clear-Sky Solar Irradiance	taub		0.299	0.339	0.364	0.432	0.396	0.391	0.417	0.439	0.386	0.363	0.336	0.305		
	taud		2.099	2.008	2.052	2.098	2.313	2.348	2.279	2.205	2.339	2.328	2.240	2.126		
	Ebn at Noon		206	231	255	252	271	273	263	248	249	228	194	179		
	Edh at Noon		23	35	41	45	38	37	39	40	31	26	21	19		
(44) All-Sky Solar Radiation	RadAvg		275	594	1071	1359	1826	1867	1857	1556	1029	518	225	152		
	RadStd		62	80	160	113	155	196	153	136	127	79	51	48		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.28	N/A	N/A	N/A	N/A	N/A	N/A	-275	+230	N/A
(47) Regional (8 neighbors)		+1.44	N/A	N/A	N/A	N/A	N/A	N/A	-319	+238	N/A

Nomenclature: See separate page